(7)

(iv) Squares of numbers from 40-60 with observation

40^2 =	16	00
41^2 =	16	81
42^2 =	17	64
43^2 =	18	49
44^2 =	19	36
45^2 =	20	25
46^2 =	21	16
47^2 =	22	09
48^2 =	23	04
49^2 =	24	01
50^2 =	25	00

50^2 =	25	00
51^2 =	26	01
52^2 =	27	04
53^2 =	28	09
54^2 =	29	16
55^2 =	30	25
56^2 =	31	36
57^2 =	32	49
58^2 =	33	64
59^2 =	34	81
60^2 =	36	00

↳ Skeleton for squares from 40 to 50 Rs

$(16+a-1) \mid (10-a)^2 = (40+a)^2$

↳ Skeleton for squares from 50 to 59 Rs

$(25+a) \mid a^2 = (50+a)^2$

(v) Square of numbers which are all 1's, 3's, 6's & 9's

1) all 1's Ex:- $11^2 = 121$

$111^2 = 12321$

$1111^2 = 1234321$

[all these logics applicable for max of 9 digits only]

logic Rs ⇒ we take 'n' digited number with all 1's then resultant Rs take naturals numbers till 'n' & then decrease till '1'.

$111111111^2 = 123456789876543 21$

(2)

2) All 3's :- Ex :- $3^2 = 09$

$33^2 = 1089$

$333^2 = 110889$

↳ Logic is that the resultant always '0' & '9' to be appeared n^{th} place & $2n^{th}$ place if we consider 'n' digited numb.

↳ here if you consider 'n' digited number resultant no. of digits are '2n' for all 3's, 6's & 9's.

↳ for all 1's :- if 'n' digited number then resultant is (2n-1) digited number.

$333^2 \rightarrow n = 3$ digits

resultant digits = $2n = 6$ digits = $--\underset{n^{th}}{0}--\underset{2n^{th}}{9}$

before digits to 'n' all are added by 'i' to '0' i.e., 0+1 = 1

before digits to '2n' all are substracted by 'i' to 9 i.e., 9-1 = 8

$333^2 = \underset{+1}{1}\ \underset{+1}{1}\ 0\ \underset{-1}{8}\ \underset{-1}{8}\ 9$

3) All 6's :- Same like all 3's but digits to be changed according to reference of $6^2 = 36$

$6^2 = \underset{\downarrow}{3}\ \underset{\downarrow}{6}$ $\quad n^{th} \rightarrow 3,\ 2n^{th} = 6$

Ex :- $6666^2 = \underset{+1}{4}\ \underset{+1}{4}\ \underset{+1}{4}\ \underset{n^{th}}{3}\ \underset{-1}{5}\ \underset{-1}{5}\ \underset{-1}{5}\ \underset{2n^{th}}{6}$

$n = 4,\ 2n = 8$

4) All 9's :- Same like all 6's but n^{th} & $2n^{th}$ digits are 8 & 1 because Ref :- $9^2 = 81$

Ex :- $99999^2 = 9\ 9\ 9\ 9\ \underset{n^{th}\ 5^{th}}{8}\ 0\ 0\ 0\ \underset{2n^{th}\ 10^{th}}{1}$

$n = 5,\ 2n = 10$

(9)

(vi) Square of any number b/w 1-100 by using base method:-

Ex:- 77^2

Step1:- for the above number we have to take base as multiple of '10' nearby '77' i.e., either 70 or 80

+7
77

70 = 7×10
77 − 70 = +7
→ Resultant

(77+7) 7 — 4 9
84×7 — $(7)^2$

5 — 8 — 4 9
+8
12

5 9 2 9

−3
77

80 = 8×10
80 − 77 = 3

(77−3) 8 — $(-3)^2$
5 9 — 2 — 9

Ans = 5 9 2 9

Step2:- take base as 70 = 7×10 (or) 80 = 8×10

Step3:- Consider the diff b/w base and given number i.e., if base 70 '+7', if base 80 '−3'.

Step4:- fill last digits with square of diff either $(+7)^2$ (or) $(-3)^2$

Step5:- given number is added (or) substracted with respective difference and multiplied with no. of 10's in base.

⇒ (77+7) 7 (or) (77−3) 8

7 → 10's in 70 ; 8 → No. of 10's in base 80.

Step6:- Result on above step is to be filled from the beginning. If 2 digits are coming @ single place, those will be added. [if >10, carry forward].

5 9 4 9
5 8 8
+1 12

(vii) Square of any 3 digit number by using $(a+b)^2$ method:-

Ex:- 234^2

Step1:- devide 234 into a & b in such a way that